A Topological Framework for Deep Learning

From a Labeled Torus to Softmax Simplex Δ_2

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TOPOLOGY DATA ANALYSIS

February 11, 2026

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Introduction

Deep learning can be understood from a topological viewpoint by interpreting a neural network as a continuous map that progressively deforms the geometry of data. Rather than viewing layers only as algebraic operations, we study how they reshape and separate regions of a space. Using a labeled torus embedded in $\mathbb{R}^3$ as a case study, we follow how successive ReLU layers transform the data until it becomes linearly separable in the probability simplex Δ_2.

Motivation

Why topology for deep learning?

- A deep network is a composition: $\mathrm{Net} = f_1 \circ \cdots \circ f_L$.
- Each layer is (piecewise) **continuous** $\Rightarrow$ the whole network is continuous.
- Topology studies continuous maps and how they deform **shapes of spaces**.

Main idea

Classification can be interpreted as **separating labeled regions** using continuous deformations.

Topological View of Data

Topological data

Assume data lies on a space X, embedded in $\mathbb{R}^{d_{in}}$. Labels form disjoint closed subsets:

$$X_L = X_1 \sqcup X_2 \sqcup \cdots \sqcup X_{d_{out}}.$$

Classification as separation

Find a continuous map $f : \mathbb{R}^{d_{in}} \to \mathbb{R}^k$ that sends each X_j into disjoint regions (classes do not overlap after mapping).

ReLU Networks = Continuous + Piecewise Linear

Typical layer

$$f(x) = \mathrm{ReLU}(Wx + b), \qquad \mathrm{ReLU}(t) = \max(t, 0).$$

Interpretation

Each layer combines:

- W: rotate / scale / shear / (possibly project),
- ReLU: fold / bend / sometimes glue parts of space.

Topological Operations

Linear part W

- Mostly topology-preserving (reorientation + scaling).
- Collapsing: projection $\Rightarrow$ quotienting.

ReLU part

- identity on positive region,
- bending/folding,
- collapse along coordinate hyperplanes (a quotient-type effect).

Dataset: Labeled Torus in $\mathbb{R}^3$

Torus parameterization

$$x = (R + r\cos\phi)\cos\theta, \quad y = (R + r\cos\phi)\sin\theta, \quad z = r\sin\phi.$$

Labels (three classes)

- **Green**: background region on torus
- **Yellow**: two separated bands (two disconnected components)
- **Purple**: band between the yellow ones ("sandwiched")

Network layers + plotted activations

Forward map

$$x \xrightarrow{W_1} z_1 \xrightarrow{\mathrm{ReLU}} a_1 \xrightarrow{W_2} z_2 \xrightarrow{\mathrm{ReLU}} a_2 \xrightarrow{W_3} z_3 \xrightarrow{\mathrm{ReLU}} a_3 \xrightarrow{W_4+b} \text{logits} \xrightarrow{\mathrm{softmax}} p$$

Plotted activations

Visualize $P0, P1, P2, P3, P4, P5, P6, P7$ (input to simplex).

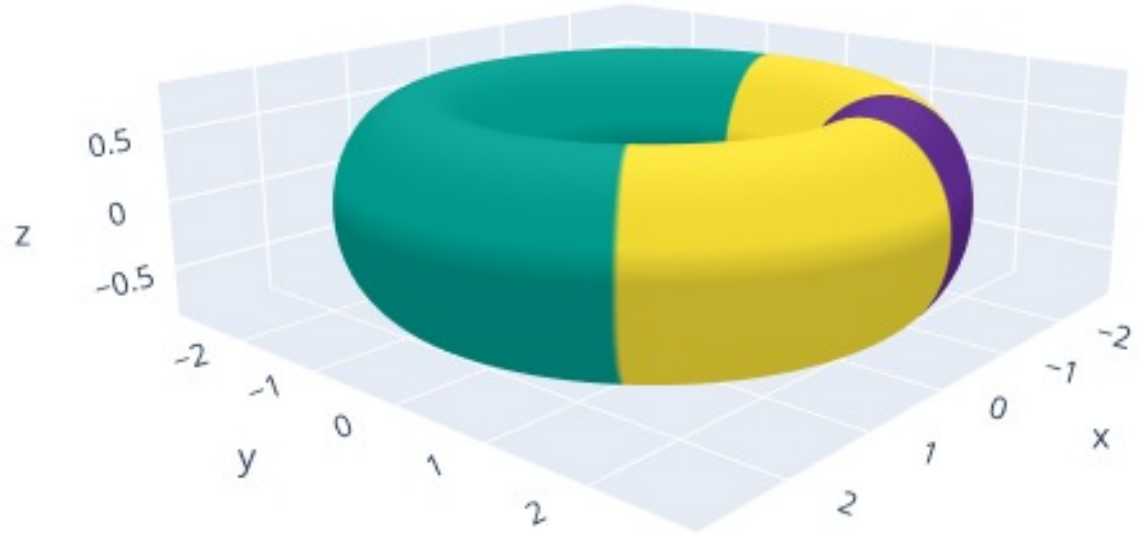
Step 1: Input space $P0$

What I start with

A torus-shaped dataset in $\mathbb{R}^3$ with three colored label regions.

Topological challenge

The **Yellow** class has two disconnected components but must be one label. So the network must "bring them together" in representation space.



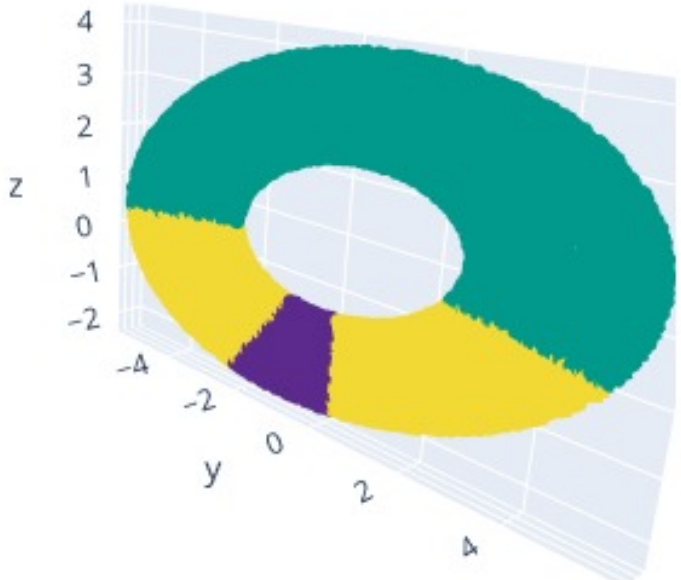
Step 2: After W_1 (projection / quotienting) — $P1$

What W_1 does

A linear map acts like a projection, collapsing one direction. This is a quotienting operation.

What training encourages

A variance penalty makes one direction small after $W_1 \Rightarrow P1$ is approximately 2D (disk-like).



Step 3: After ReLU_1 (bending / folding) — $P2$

What ReLU does

$\mathrm{ReLU}(t) = \max(t, 0)$ folds space at coordinate hyperplanes (negative values become 0).

Topological meaning

ReLU can bend parts of the space and can also identify/collapse parts (quotient-type effect), improving separability.

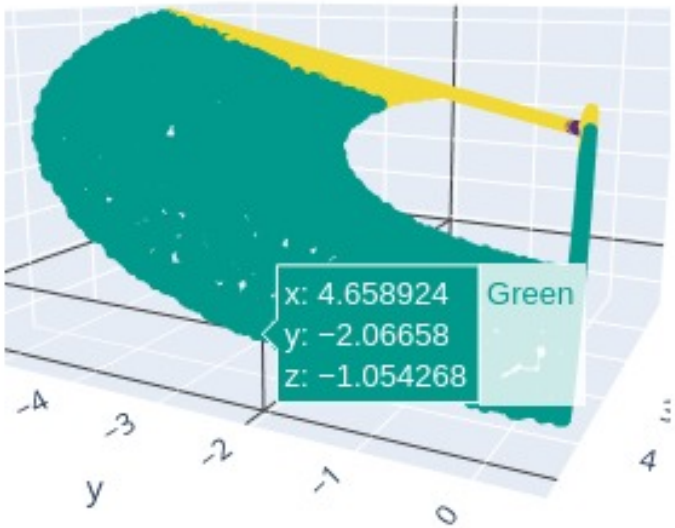
Step 4: After W_2 (reorientation) — $P3$

What W_2 does

A linear map rotates/reorients the cloud so that directions useful for separating classes are aligned.

What training encourages

An orthogonality penalty pushes $W_2^T W_2 \approx I$, so W_2 behaves like a near-rotation.



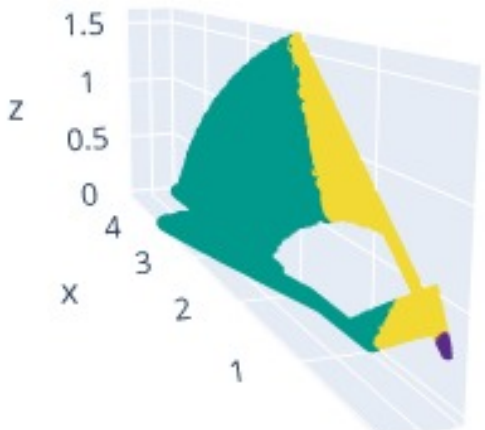
Step 5: After ReLU_2 (more bending) — $P4$

Effect

After reorientation, ReLU folds again. This can move the two **Yellow** components closer in representation space.

Why it matters

To classify correctly, both **Yellow** components must land in the same output region. ReLU + projections can help identify them.



Step 6: After W_3 (controlled collapse) — $P5$

What W_3 does

Another linear map compresses the representation while preserving what is needed to separate labels (quotienting if strong collapse occurs).

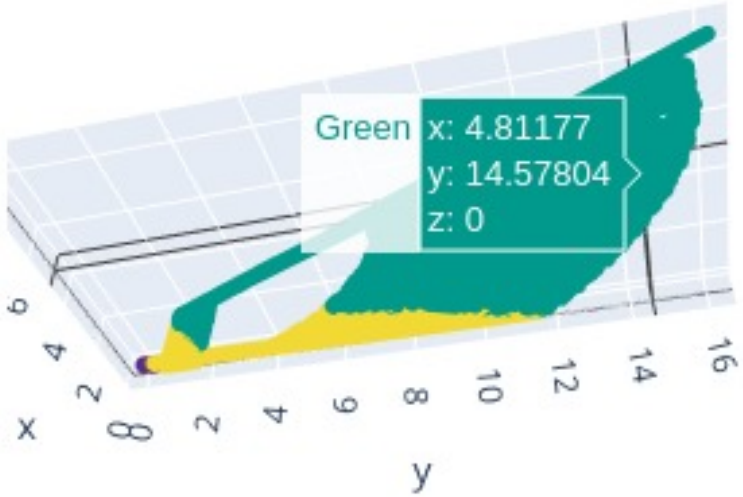
Figure: Activation after W_3.

Step 7: After ReLU_3 (final alignment) — $P6$

Effect

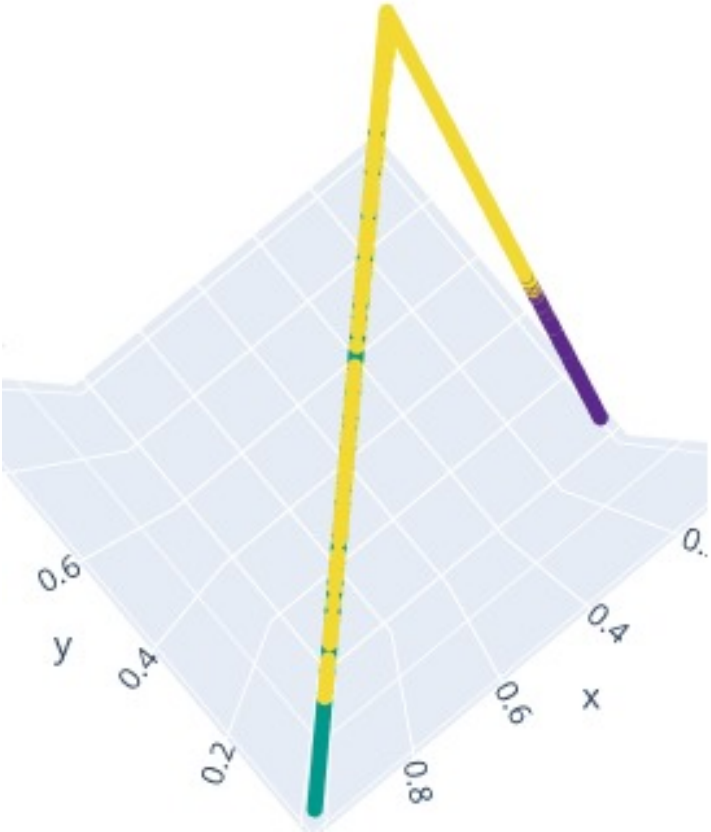
ReLU bends the compressed representation and helps position each class before the final linear layer.

This brings **Yellow** points closer and preserve separation from **Purple** and **Green**.



Step 8: Softmax output $P7 \in \Delta_2$

$$\mathrm{softmax}(u)_i = \frac{e^{u_i}}{e^{u_1} + e^{u_2} + e^{u_3}}, \qquad p_i \geq 0, \ \sum_i p_i = 1 \Rightarrow p \in \Delta_2.$$



Conclusion

Main takeaways

- Classification can be seen as topological separation.
- ReLU networks act by finite operations: rotate/scale/project + bend/glue.
- Softmax maps to simplex Δ_{n-1} (Δ_2).

Thank you.

Questions?

Reference

Hajij, M., & Istvan, K. (2020). *A topological framework for deep learning*. arXiv preprint arXiv:2008.13697.